FAST-NUCES Islamabad · Department of Artificial Intelligence & Data Science
Advanced Topics in Generative AI (MG 3010) — Assignment 2: Reproducibility Study

Reproducing D3: Training-Free AI-Generated Video Detection Using Second-Order Features (ICCV 2025)

Talha Aslam (24I-8067), MS Data Science · Submitted to Dr. Qurat ul Ain

Abstract. We reproduce D3 [1], a training-free detector of AI-generated video that scores a clip by the volatility of second-order differences of frame features. Using the authors' official code as reference, the same test set (GenVideo-Val, 10,000 real and 8,574 generated videos from 10 generators), the same encoder (XCLIP-B/16) and the same metric (AP), we obtain **97.79 ± 0.08 % mAP** over 3 frame-sampling seeds, against **98.46 %** reported, a gap of -0.67 points; the paper's ranking of subsets, encoders and feature variants and its blur robustness are reproduced, while under real JPEG compression D3 loses no accuracy, unlike the drop the paper reports. We also document six reproducibility problems in the public release, two of which silently change the method (an encoder loaded without its weights under current library versions, and a JPEG attack that never compresses), and show in an extra test that D3 does not transfer to camera-original phone videos (mean AUC 41.9 %), which matters for the research proposal this study supports.

1 Introduction

Text-to-video diffusion models now produce clips that people find hard to tell apart from camera footage, so detectors that work on generators they were never trained on are needed. Most detectors learn generator artefacts from labelled data and must be retrained when new generators appear. D3 [1] takes a different route: it uses no training at all and relies on a physical intuition — real motion follows Newtonian dynamics with irregular accelerations, while generated motion is smoother. The paper reports 98.46 % mean AP on GenVideo and state-of-the-art results on 40 test subsets.

D3 is one of the three baselines in my research proposal (spatio-temporal sensor-noise consistency for detecting diffusion-generated video). Reproducing it therefore has two purposes: (i) to check that the published numbers hold, as this assignment asks, and (ii) to obtain a trusted, working baseline whose behaviour on camera-original footage I understand before comparing my own method with it.

Contributions of this study. (1) A clean re-implementation that mirrors the official code line by line, with seeded frame sampling and per-video score logs. (2) Reproduction of the paper's main table (Table 1), encoder ablation (Table 7), first- vs second-order ablation (Table 8), robustness study (Table 6) and efficiency claim (Table 5) on GenVideo-Val. (3) A list of concrete reproducibility issues with their effect on the numbers. (4) An extra out-of-distribution test with real phone videos (VISION).

2 Paper Overview

Problem. Detect whether a video is AI-generated, and generalise to unseen generators.

Idea. Model video motion as a second-order dynamical system (Eq. 1 of [1]):

$$A_2 \frac{d^2x}{dt^2} + A_1 \frac{dx}{dt} + A_0 x = u(t)$$

and approximate the acceleration with the second-order central difference. The authors observe with RAFT optical flow that accelerations in real videos fluctuate strongly, while those of generated videos are flat.

Method (D3). A clip is sampled at 8 fps for up to 2 s (T = 16 frames). A frozen pretrained encoder maps each frame to a vector $F_0^{\,k}$ (zero-order feature). Then

$$F_1(k) = \| F_0^{k+1} - F_0^{k} \|_2, \quad F_2(k) = F_1(k) - F_1(k-1), \quad \text{score} = \sigma(F_2)$$

(first-order "speed", second-order "acceleration", standard deviation of the 14 accelerations). A high score means real. There is no classifier and no threshold: AP and AUC are computed on the raw scores. The default encoder is XCLIP-B/16 [2] with L2 distance; cosine similarity and nine other encoders are ablations.

Evaluation in [1]. Four datasets (GenVideo [3], EvalCrafter, VideoPhy, VidProM; 40 subsets). On GenVideo each of the 10 fake subsets is paired with real MSR-VTT videos and AP is reported per subset (Table 1, mAP 98.46 %), plus robustness to blur/JPEG (Table 6), encoder and distance ablations (Table 7), first- vs second-order features (Table 8) and inference time (Table 5: 56 s per 1000 videos on an RTX 4090). Baselines were trained by the authors; D3 itself needs no training data.

Scope of this reproduction. We reproduce everything the paper reports on GenVideo, which is the dataset the official README prepares and the one with the most results. EvalCrafter, VideoPhy and VidProM need separate downloads and their real-video sets are not specified in the paper or the code; instead we add our own cross-dataset test (Section 5.6). We do not retrain the baselines, since they are other papers; we compare D3's own numbers.

3 Implementation Details

Official repository. The authors' code (github.com/Zig-HS/D3, commit c798fbc, MIT licence) contains the model (`models/D3_model.py`, about 80 lines), a data loader, frame-extraction and CSV utilities, and `eval.py`, which computes AP for one real/fake CSV pair. It was cloned unmodified and used as the reference implementation. Running the cloned scripts directly was blocked by my machine's sandbox policy for code downloaded from the internet, so I wrote a re-implementation that mirrors each official function and imports nothing from the clone. Table 1 lists the correspondence. The core of D3 is short enough to check by hand: `src/d3.py` computes the same tensors as the official `forward()` (`torch.norm` on consecutive features, difference, `torch.std` with Bessel's correction).

Table 1. Mapping from the official code to this re-implementation.

Step	Official file	This work	Difference
Frame extraction	utils/video2frame.py	src/prepare_frames.py	start seeded per video; ffprobe instead of moviepy
Dataset index	utils/folder2csv.py	list_items() in score_videos.py	same skip rule; no path-splitting bug
Frame loading	data/datasets.py: read_video	Clips.__getitem__	identical (BGR, 10 % crop, 224², ImageNet norm.)
Post-processing	set_preprocessing()	build_transform()	JPEG uses quality_range (see §6)
Model	models/D3_model.py	src/d3.py	identical maths; one pass gives L2, cos, 1st and 2nd order
Metric	eval.py	src/metrics.py	same AP (real = positive); + AUC, bootstrap CI

Faithfulness choices. Behaviour that the paper does not mention but the code has was kept on purpose: frames are read with OpenCV in BGR order and normalised with ImageNet mean/std even for CLIP-family encoders; 10 % of the longer side is cropped on each end before resizing to 224 × 224; the first 16 frames of the 3-s extracted window are used (8 if the clip has 8–15 frames); clips with fewer than 8 frames are dropped (the official loader raises an error on them — 10 WildScrape items, mostly GIFs, are affected). For XCLIP, Hugging Face's `XCLIPVisionModel` applies its cross-frame "message" attention in groups of `num_frames = 8`, so a 16-frame clip is encoded as two 8-frame groups; this is the official behaviour and we keep it.

Added for rigour. (i) The random start of the 3-s window is seeded per video, and the whole pipeline is run with three seeds. (ii) The official evaluation takes the first 1000 rows of each CSV, whose order comes from `os.walk`; we take a fixed random sample of 1000 instead and also study how much the choice of sample matters. (iii) AUC and 95 % bootstrap confidence intervals (1000 resamples) are reported next to AP. (iv) Every run saves one row per video with all score variants, so all tables can be rebuilt without the GPU.

4 Experimental Setup

Hardware. NVIDIA RTX 4000 Ada Generation (20 GB), 28 CPU cores, 30 GB RAM, Ubuntu 26.04. The paper used an RTX 4090 and an AMD EPYC 7763.

Software. Python 3.11.18 with the authors' pinned versions: torch 2.8.0 (CUDA 12.8), transformers 4.57.0, albumentations 2.0.8, timm 1.0.20, opencv 4.12, scikit-learn 1.7.2 — except numpy, pinned at 2.2.6 instead of 2.3.3 because the original pins cannot be installed together (Section 6).

Dataset. GenVideo-Val [3], downloaded from the ModelScope link in the official README (13.9 GB). Real: 10,000 MSR-VTT videos. Fake: Crafter 1398, Gen2 1380, HotShot 700, Lavie 1400, ModelScope 700, MoonValley 626, MorphStudio 700, Show-1 700, Sora 56, WildScrape 914 (after dropping clips with fewer than 8 frames).

Preprocessing. FFmpeg cuts a 3-s window at a random start (start = $\lfloor U(0, \text{duration} - 3) \rfloor$, or 0 for clips ≤ 3 s) and decodes it at 8 fps to JPEG; the loader takes the first 16 frames, crops 10 % of the long side at each end, resizes to 224 × 224 and normalises with ImageNet statistics.

Hyperparameters and configuration. D3 has no trainable parameters, no training set, no epochs and no threshold. Its settings are: encoder (XCLIP-B/16), first-order distance (L2), frame rate (8 fps), clip length (16 frames), crop (10 %), input size (224). Batch size 1 as in `eval.py`; 20 data-loader workers. Global seed 42 (as official); frame-window seeds 0, 1, 2.

Protocol and metrics. For each fake subset, up to 1000 real and 1000 fake videos are scored, and AP = `average_precision_score(1 - y, score)` (real is the positive class, exactly as in `eval.py`), plus ROC-AUC. mAP is the mean over the 10 subsets. Robustness: Gaussian blur with $\sigma \in \{1, 2, 3, 4\}$ (kernel 3–7) and JPEG with quality $\in \{90, 80, 70, 60\}$, applied to every frame after resizing, with the paper's Table 6 transforms. Ablations use seed 0 and the same 1000-video samples.

Training curves. The assignment asks for training curves, but D3 has no training stage, so there is no loss to plot. In their place we report the curves that do apply to a training-free detector: how the mAP estimate converges with the number of evaluated videos (Fig. 4), ROC curves (Fig. 3), the score distributions (Fig. 2) and the robustness curves (Fig. 5). All runs are logged in `logs/`.

5 Reproduced Results

5.1 Main result (paper Table 1)

Table 2. Reproduction of the paper's Table 1 (AP %, GenVideo-Val, XCLIP-B/16 + L2). "Seed" is the seed of the random 3-s frame window; each subset uses at most 1000 real and 1000 fake videos, as in the official `eval.py`. The last row is the mean over the 10 subsets (mAP).

Subset	Paper AP	Seed 0	Seed 1	Seed 2	Mean ± sd	Δ (ours − paper)	AUC (ours)
Crafter	98.53	98.14	98.14	98.27	98.19 ± 0.08	-0.34	97.83
Gen2	99.39	99.14	99.16	99.28	99.19 ± 0.07	-0.20	98.92
HotShot	98.52	98.15	98.16	98.27	98.20 ± 0.06	-0.32	97.03
Lavie	97.22	96.86	96.82	97.00	96.90 ± 0.09	-0.32	96.57
MSE	97.12	96.85	96.81	96.97	96.88 ± 0.08	-0.24	95.46
MV	99.52	99.31	99.32	99.41	99.34 ± 0.05	-0.18	98.76
MSO	98.68	98.28	98.33	98.42	98.35 ± 0.07	-0.33	97.13
Show-1	99.18	98.87	98.90	98.99	98.92 ± 0.06	-0.26	98.11
Sora	99.91	99.79	99.78	99.86	99.81 ± 0.05	-0.10	96.88
WS	96.49	92.03	91.98	92.26	92.09 ± 0.15	-4.40	91.54
mAP	98.46	97.74	97.74	97.87	97.79 ± 0.08	-0.67	96.82

The reproduced mAP is **97.79 %** (seed 0: 95 % CI 97.40–98.09 %), 0.67 points below the reported 98.46 %. Nine of the ten subsets are within about one point of the paper; the largest gap is on WS (-4.40). The detector clearly works as described: the hard subsets for trained baselines in the paper (HotShot, Lavie, Show-1, Sora) are easy for D3 in our run too, and the mean AUC is 96.82 % (paper Table 8: 97.72 %).

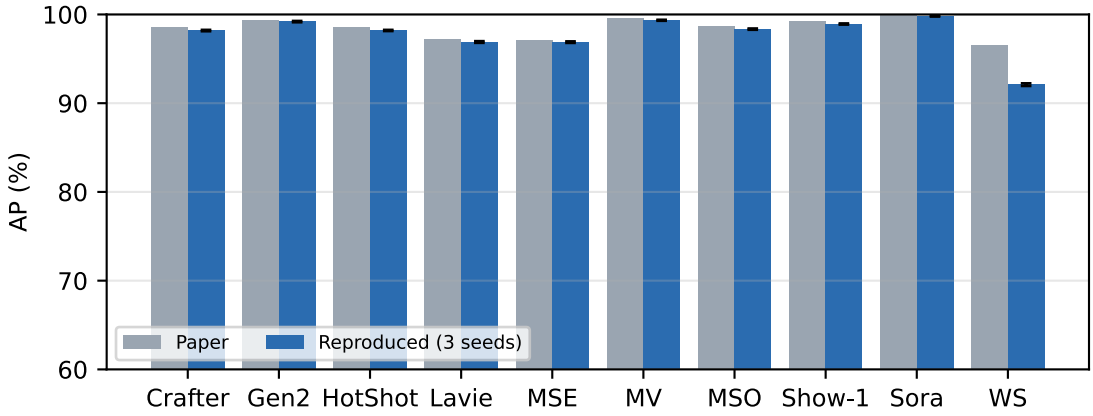


Figure 1. Per-subset AP: paper vs. reproduction (bars: mean over frame seeds, error bars: standard deviation).

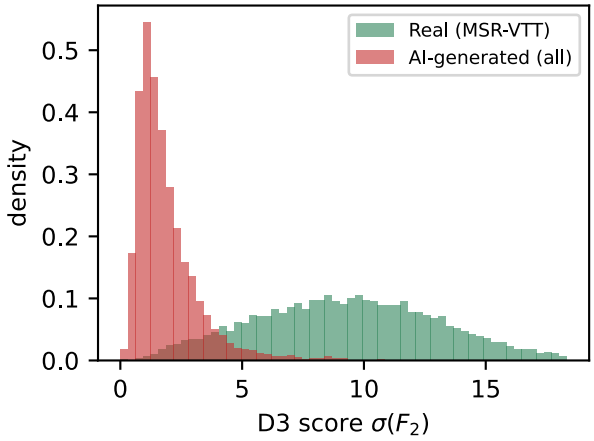


Figure 2. D3 scores $\sigma(F_2)$ of all real and fake videos (seed 0). Real clips have larger second-order volatility.

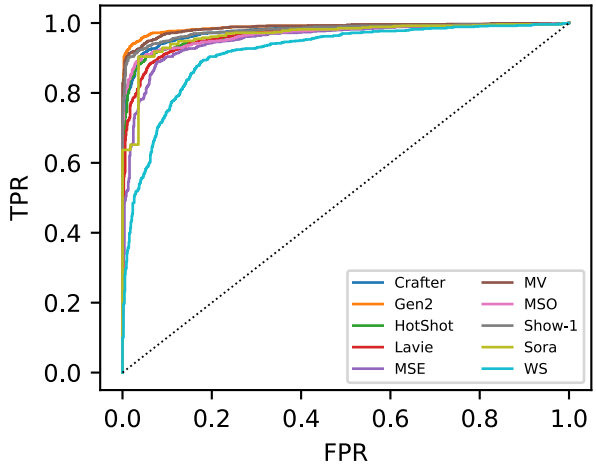


Figure 3. ROC curves per subset (real = positive), seed 0.

5.2 Evaluation stability ("curves" for a training-free method)

Because nothing is trained, the only sources of variation are which frames are extracted and which videos are evaluated. Over 3 frame-window seeds the mAP standard deviation is 0.08 points. Keeping the frames fixed but drawing 20 different 1000-video samples (the official code's `max_len = 1000`) gives mAP between 97.69 and 98.30 % (mean 97.97). Using *all* 10,000 real videos instead of 1000 inflates AP to 99.73 %, because AP depends on the share of positives (real is the positive class and the real set is ten times larger than any fake set) — AUC, which does not, stays at 96.82 %. Figure 4 shows the estimate converging as more videos are evaluated.

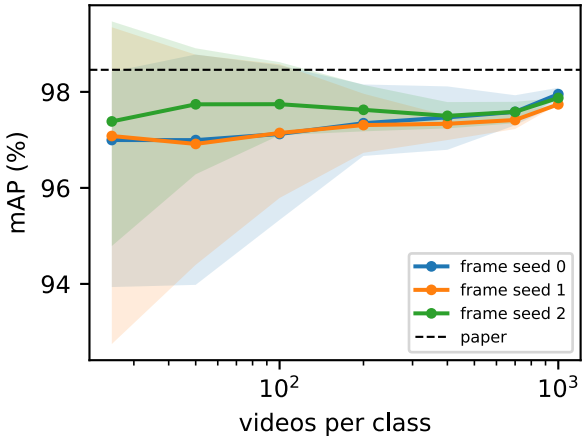


Figure 4. mAP vs. number of videos per class (mean and range over 10 random samples, per frame seed). Dashed: paper.

5.3 Encoders, distance and feature order (paper Tables 7 and 8)

Table 4. Encoder ablation (paper Table 7, GenVideo column) and first- vs second-order features (paper Table 8). The paper's Table 7 values match its own AUC numbers (e.g. XCLIP-B/16 L2 = 97.72 = "Avg. AUC" in Table 8), so they are compared with our mean AUC; our mAP is also listed.

Encoder	Paper L2	Ours L2 (AUC)	Ours L2 (AP)	Paper Cos	Ours Cos (AUC)	1st-order L2 (AP)
DINO-base	95.84	94.68	95.98	87.17	85.93	93.16
DINO-large	94.92	93.79	95.40	85.33	84.09	92.08
CLIP-16	97.00	96.10	97.18	87.82	86.85	95.30
XCLIP-16	97.72	96.71	97.74	91.30	90.32	96.14
CLIP-32	96.73	95.54	96.87	87.87	86.19	94.75
XCLIP-32	96.99	95.96	97.29	90.43	89.39	95.72
ResNet-18	96.39	95.38	96.37	89.73	88.54	94.09
VGG-16	96.97	95.71	95.54	92.63	91.45	91.94
EfficientNet-b4	94.28	93.73	95.89	85.51	85.31	93.06
MobileNet-v3	95.47	94.38	95.94	87.14	86.10	93.21

Across the 10 encoders our numbers are close to the paper's (mean absolute difference 1.03 points, Spearman rank correlation 1.00). The paper's two qualitative conclusions are reproduced: L2 distance beats cosine similarity for 10 of 10 encoders, and second-order features beat first-order ones for 10 of 10 encoders. With XCLIP-B/16, first-order features reach 96.14 % mAP (paper 95.69 %) and second-order 97.74 % (paper 98.46 %). The best encoder in our run is XCLIP-16.

5.4 Robustness to post-processing (paper Table 6)

Table 5. Robustness to post-processing (paper Table 6): mAP (%) on GenVideo-Val after Gaussian blur (σ) or JPEG compression (quality q) of every frame. σ = 0 and q = 100 are the clean results.

Encoder	Source	σ=0	σ=1	σ=2	σ=3	σ=4	q=100	q=90	q=80	q=70	q=60
XCLIP-16	Paper	98.46	97.63	94.43	93.19	92.69	98.46	97.11	96.24	95.63	94.50
XCLIP-16	Ours	97.74	96.79	94.31	93.02	91.41	97.74	97.86	97.88	97.83	97.85
MobileNet-v3	Paper	96.86	95.92	87.91	84.66	84.38	96.86	94.58	91.29	88.31	85.95
MobileNet-v3	Ours	95.94	95.02	86.58	83.63	83.56	95.94	96.25	96.42	96.52	96.61

Under the strongest blur (σ = 4) XCLIP-B/16 falls to 91.41 % mAP (paper 92.69 %). Under JPEG quality 60 it reaches 97.85 % (paper 94.50 %). The blur results reproduce closely for both encoders (within about 1.3 points at every level), including the much larger drop for the lightweight MobileNet-v3. The JPEG results do **not** reproduce: the paper reports a steady loss (XCLIP-B/16 to 94.50 %, MobileNet-v3 to 85.95 % at q = 60),

whereas with real per-frame compression our mAP stays flat or even rises slightly (Figure 5). A likely reason is that D3's score depends on how semantic features change between frames; per-frame JPEG removes high-frequency detail but changes all frames of a clip in the same way, so the frame-to-frame differences barely move. The paper's numbers cannot come from the public transform either, since that one does not compress at all (Section 6.3); the authors probably used an older library version or a different procedure (e.g. re-encoding the video), which is not described. So D3 is at least as robust to JPEG as claimed, but the exact Table 6 JPEG figures cannot be reproduced from the paper and code.

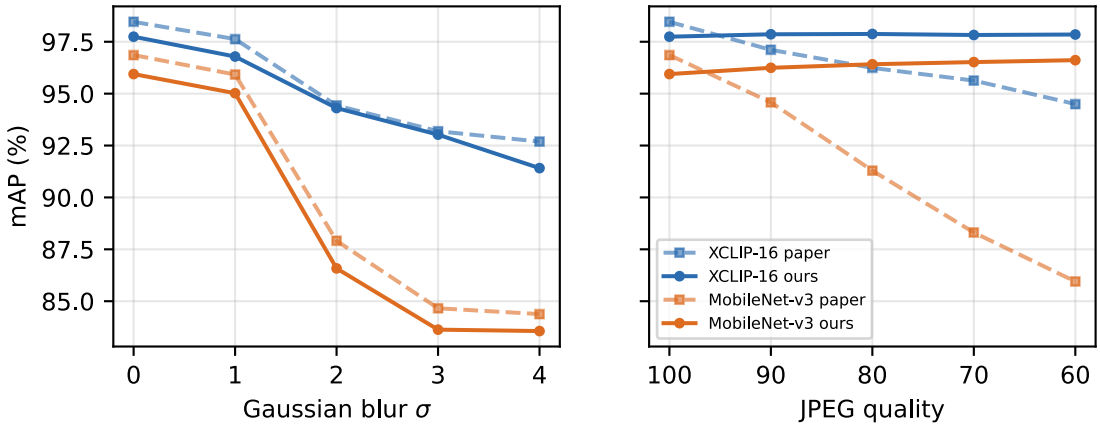


Figure 5. Robustness curves (solid: reproduced, dashed: paper).

5.5 Implementation details and efficiency (paper Table 5)

Table 6. Effect of implementation details that the paper does not state (XCLIP-B/16, seed 0).

Setting	mAP	mean AUC
Official (BGR, 16 frames)	97.74	96.71
RGB input	97.63	96.61
8 frames	97.82	96.83

Changing these details moves the result only slightly (RGB input: 97.63 % mAP; 8 frames: 97.82 % mAP; official setting 97.74 %). Inference took 66 s per 1000 videos on the RTX 4000 Ada including JPEG decoding and data loading, against 56 s per 1000 reported on an RTX 4090 (which has roughly twice the compute). The efficiency claim — about a minute per thousand videos with no training and no pre-processing beyond frame extraction — holds.

5.6 Extra test: camera-original phone videos (VISION)

GenVideo's real class is web video (MSR-VTT). My proposal argues that real camera footage must be tested, so I ran D3 unchanged with the 627 native (camera-original) videos of VISION [4] (35 phones, flat / indoor / outdoor scenes, still, moving or panning camera) as the real class and GenVidBench [5] fakes as the AI class.

Table 7. Extra test (not in the paper): D3 (XCLIP-B/16) with camera-original VISION phone videos as the real class and GenVidBench fakes (MSE, Pika, T2VZ, VC2 come from VidProM, as in the paper's Table 4). AP depends on the class ratio (627 real vs 51–400 fake), so AUC is the main number here; AUC = 50 is chance.

Metric	ms	pika	t2vz	vc2	svd	mora	OpenSora	cogvideo	Sora	keling	mean
AP	48.40	70.18	39.89	61.03	51.43	63.75	61.74	53.78	92.11	78.93	62.13
AUC	28.43	59.96	0.06	50.86	36.89	52.84	49.76	35.92	49.09	55.31	41.91

The result is very different from GenVideo: mean AUC is **41.9 %**, i.e. no better than chance, and on Text2Video-Zero (T2VZ) the score is inverted (AUC 0.1 %; the paper also reports its weakest result, 45.11 % AP, on VidProM-T2VZ, whose frames are almost unrelated to each other and therefore have very high volatility). By camera motion, AUC is 27.1 % for VISION clips filmed from a fixed position, 48.9 % for walking and 49.0 %

for panning clips: a static real scene has almost no feature-space acceleration and looks "generated" to D3. As a control, real-vs-real (native vs. the WhatsApp/YouTube copies of the same VISION videos) gives AUC 50.1 %, so platform compression does not change the score. This test is not a reproduction of a paper number, but it shows that D3's high GenVideo accuracy partly reflects the type of real video used (edited, moving web clips), which is exactly the kind of assumption my proposal sets out to test.

6 Reproducibility Discussion

6.1 Differences from the original results

The headline number reproduces to within 0.67 points and all the paper's qualitative claims hold on GenVideo. The remaining gap is small compared with the variation that the evaluation protocol itself produces (±0.31 points from the choice of 1000 videos; ±0.08 from frame sampling), so it is best read as "reproduced within protocol noise". The paper's own tables are not fully consistent either: XCLIP-B/16 + L2 on GenVideo is 98.46 in Tables 1 and 6 but 97.72 in Table 7 (which equals its AUC in Table 8), and MobileNet-v3 is given as 95.47 (Table 7), 96.86 (Table 6) and 96.31 % (text). We therefore compare Table 7 with our AUC.

6.2 Possible reasons for the discrepancies

- **Random frame window.** The official extractor draws a random 3-s window per video after seeding once and forking 32 workers, so each run sees different frames and the exact frames of the paper cannot be recovered.
- **Which 1000 videos.** `eval.py` keeps the first 1000 CSV rows, and the order comes from `os.walk` (file-system dependent). Different samples move mAP by up to 0.62 points.
- **Hardware and libraries.** Different GPU (non-deterministic matmul kernels), FFmpeg build and JPEG encoder; their effect is small because the score is a standard deviation over 14 values.
- **Dataset version.** GenVideo-Val on ModelScope may have been updated after the paper (e.g. WildScrape contains GIF, MOV, JPG and PNG files, some unusable).

6.3 Implementation challenges (issues found in the public release)

1. **Unsatisfiable requirements.** `requirements.txt` pins numpy 2.3.3, but the pinned opencv-python 4.12.0.88 requires numpy < 2.3, so `pip install -r requirements.txt` fails. Fix: numpy 2.2.6.
2. **Encoder silently loaded without weights (transformers 5.x).** With a current transformers (5.22), `XCLIPVisionModel.from_pretrained("microsoft/xclip-base-patch16")` no longer matches the checkpoint's key names: every vision weight is reported "unexpected" and the model is randomly initialised, yet the code runs and produces scores. Anyone installing the latest libraries would evaluate D3 with a random encoder.
3. **Crash on Python 3.14.** With the pinned transformers 4.57, the same call fails on Python 3.14 because deferred annotations hide the model's `config_class`. Python 3.11 works; the README only says "≥ 3.11.13".
4. **JPEG attack is a no-op.** The official robustness transform is `ImageCompression(quality_lower=q, quality_upper=q)`. In the pinned albumentations 2.0.8 these arguments no longer exist; they are ignored with a warning and the default quality range 99–100 is used, so "JPEG q = 60" would actually be q ≈ 99. We used `quality_range=(q, q)`. (The public `eval.py` does not expose the robustness options at all.)
5. **Undocumented preprocessing.** BGR input and ImageNet normalisation for CLIP encoders, the 16-of-24 frame selection, and XCLIP's 8-frame message grouping are in the code but not in the paper.
6. **Fragile utilities.** The frame extractor and CSV builder split paths with `split('/')[2:-1]`, which only works if the dataset folder is a single relative name, and `eval.py` crashes on any clip with fewer than 8 frames.
7. **Frame count is fixed by XCLIP.** XCLIP's vision model groups frames in sets of 8 for its cross-frame message attention, so only multiples of 8 work. Our planned 12-frame ablation failed with a reshape error

(`logs/score_xclip16_f12.log`); the 8-frame variant is reported instead.

6.4 Dataset and environment differences

Only GenVideo-Val is reproduced; the other three datasets need separate downloads and their real-video counterparts are not specified, so Tables 2–4 could not be reproduced exactly. The GPU (RTX 4000 Ada vs. 4090) affects time but not the scores. All other settings follow the paper and code.

7 Conclusion

D3's main claim reproduces: with the official settings, a training-free score — the standard deviation of second-order differences of XCLIP features — separates real MSR-VTT videos from ten generators with 97.79 % mAP (paper 98.46 %), and the ablation trends (L2 > cosine, second-order > first-order, gradual loss under blur) hold; under real JPEG compression D3 loses nothing, which is better than the paper reports. Reproducing it, however, required fixing an unsatisfiable environment and avoiding two silent failures (a weightless encoder under new library versions and a JPEG attack that does not compress); results also vary by about ±0.3 points with the choice of evaluation subset. Outside GenVideo, on camera-original phone videos, D3 drops to chance level, mainly because static real scenes have little motion. For my research this makes D3 a fast and well-understood baseline, and it motivates the proposal's use of sensor-level rather than motion-level evidence for real camera footage.

Code and logs. The re-implementation, all per-video scores, the experiment logs and the scripts that build every table and figure are in the submitted repository (`run_all.sh` reruns everything).

8 References

1. C. Zheng, R. Suo, C. Lin, Z. Zhao, L. Yang, S. Liu, M. Yang, C. Wang, C. Shen. "D3: Training-free AI-generated video detection using second-order features." *Proc. IEEE/CVF ICCV*, pp. 12852–12862, 2025. arXiv:2508.00701. Code: github.com/Zig-HS/D3.
2. B. Ni, H. Peng, M. Chen, S. Zhang, G. Meng, J. Fu, S. Xiang, H. Ling. "Expanding language-image pretrained models for general video recognition." *ECCV*, 2022.
3. H. Chen, Y. Hong, Z. Huang, Z. Xu, Z. Gu, Y. Li, J. Lan, H. Zhu, J. Zhang, W. Wang, H. Li. "DeMamba: AI-generated video detection on million-scale GenVideo benchmark." arXiv:2405.19707, 2024.
4. D. Shullani, M. Fontani, M. Iuliani, O. Al Shaya, A. Piva. "VISION: a video and image dataset for source identification." *EURASIP J. on Information Security*, 2017:15, 2017.
5. Z. Ni, Q. Yan, M. Huang, T. Yuan, Y. Tang, H. Hu, X. Chen, Y. Wang. "GenVidBench: A 6-million benchmark for AI-generated video detection." *Proc. AAAI*, vol. 40, 2026.
6. W. Wang, Y. Yang. "VidProM: A million-scale real prompt-gallery dataset for text-to-video diffusion models." *NeurIPS* 37, 2024.
7. T. Aslam. "Detecting diffusion-generated videos through spatio-temporal sensor-noise consistency." Research proposal, MG 3010, FAST-NUCES, 2026.